

Project Design Phase-II

Data Flow Diagram & User Stories

Date	13 July 2026
Team ID	
Project Name	Voice-Based Concept Understanding Analyzer (VBCUA)
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD depicts the right amount of system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

The diagrams below describe the data flow of the Voice-Based Concept Understanding Analyzer (VBCUA): Figure 1 presents the Level 0 (context) view of the system as a single process exchanging data with its user, and Figure 2 decomposes that process into the Level 1 pipeline of transcription, semantic comparison, scoring, and report generation implemented in the application.

DFD Level 0 (Context Diagram) — VBCUA



Figure 1: DFD Level 0 (Context Diagram) — VBCUA

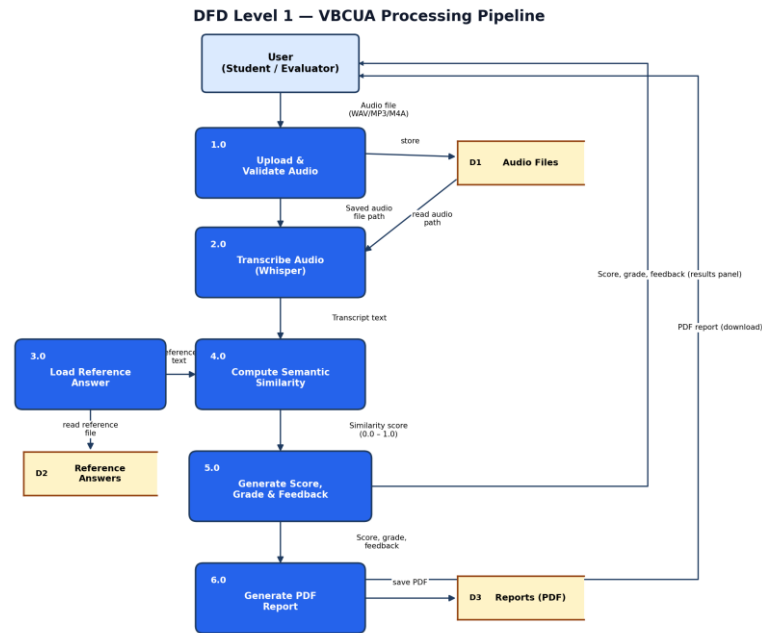


Figure 2: DFD Level 1 — VBCUA Processing Pipeline

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Student (End User)	Audio Upload	USN-1	As a student, I can upload a recorded audio file of my spoken explanation so that it can be evaluated.	Selecting a WAV, MP3, or M4A file uploads successfully and a confirmation message with the file name is shown.	High	Sprint-1
		USN-2	As a student, I want unsupported audio file types to be rejected so that I only upload a file the system can process.	Selecting a file outside the WAV/MP3/M4A formats is blocked by the uploader before submission.	Medium	Sprint-1
	Speech-to-Text Transcription	USN-3	As a student, I want my audio automatically transcribed into text so that I can see what the system understood from my explanation.	After upload, a text transcript of the spoken response is displayed on screen.	High	Sprint-1
		USN-4	As a student, I want to be notified if transcription fails so that I know to re-upload my audio.	A descriptive error message is shown when transcription cannot be completed.	Medium	Sprint-2
	Semantic Similarity Analysis	USN-5	As a student, I want my transcript compared against a reference	A similarity score between 0% and 100% is generated	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
			answer so that my understanding can be measured objectively.	once transcription completes.		
	Concept Understanding Scoring	USN-6	As a student, I want to receive a grade (Excellent / Good / Fair / Needs Improvement) so that I can quickly gauge my performance.	A grade label is displayed together with the percentage score.	High	Sprint-1
		USN-7	As a student, I want written feedback explaining my grade so that I understand what to improve.	A feedback message specific to the grade tier is shown below the score.	Medium	Sprint-1
	Results Display	USN-8	As a student, I want to view my transcript, score, and feedback together on one screen so that I can review my performance at a glance.	The results panel displays transcript, similarity metric, grade badge, and feedback simultaneously.	Medium	Sprint-1
	PDF Report Generation	USN-9	As a student, I want a PDF report of my assessment generated automatically so that I can keep a record of my performance.	After analysis completes, a PDF containing the transcript, score, grade, and feedback is created.	Medium	Sprint-2
	Report Download	USN-10	As a student, I want to download my PDF report so that I can save or share it.	A working download button appears once a report has been generated for the current session.	Medium	Sprint-2
Evaluator / Instructor	Reference Answer Management	USN-11	As an evaluator, I want the system to use a stored reference answer for comparison so that grading is based on a consistent standard.	The system loads and validates the reference answer file before scoring begins.	High	Sprint-1
	Results Display	USN-12	As an evaluator, I want each submission to show a clear grade tier so that I can quickly identify students who need support.	Each result includes a colour-coded grade badge corresponding to its assessment tier.	Medium	Sprint-2